

ABEL DOÑATE MUÑOZ

Mathematics and Engineering Physics

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📍 Bonn, Germany

EDUCATION

Master of Mathematics

University of Bonn

📅 2025 – 📍 Bonn, Germany

Student in CFIS program

Polytechnic University of Catalonia (UPC)

📅 2020 – 2025 📍 Barcelona, Spain

CFIS is a special excellence school created by UPC university to host some of the best mathematicians, physics and engineering students in Spain, offering them the possibility of making a double degree.

Degree on Mathematics

Polytechnic University of Catalonia (UPC)

📅 2020 – 2024 📍 Barcelona, Spain

Degree on Engineering Physics

Polytechnic University of Catalonia (UPC)

📅 2020 – 2024 📍 Barcelona, Spain

Development and defense of Senior thesis ([see](#))

University of Utah

📅 2024 – 2025 📍 Salt Lake City

EXPERIENCE

Participation in seminar of Higher Category Theory

Graduate seminars

📅 2025 📍 University of Bonn

Participation in the seminar organized by students of Higher Category Theory

Participation in different seminars

Graduate seminars BIKES and BAGELS

📅 2024 📍 University of Utah

Participation in the seminars organized by students of commutative algebra (BIKES) and algebraic geometry (BAGELS). Talk in BIKES about Bernstein-Sato theory in positive characteristic.

FURTHER EXPERIENCE

Research stay in ICMAT

Institute of Mathematical Sciences

📅 2023, July 📍 Autonomous University of Madrid

Research on Bounded cohomology of groups and quasi-morphisms during a stay in ICMAT (Instituto de Ciencias Matemáticas) with a JAE-INTRO scholarship. (lecture [here](#)).

Research stay in CRM

Mathematical Research Centre

📅 2022, July 📍 University of Barcelona

Research on Coxeter Matroids during a stay in CRM (Centre de Recerca Matemàtica) during Barcelona Introduction to Mathematical Research (lecture [here](#)).

Readings on Connective Constant in Regular Lattices

Polytechnic University of Catalonia

📅 2023 📍 Barcelona

Reading of papers and development of techniques guided by a professor of Polytechnic University of Catalonia on the connective constant of regular lattices. Main paper to understand [here](#).

Fellow intern of the subject of computer science

Polytechnic University of Catalonia

📅 2022 – 2023 📍 Barcelona

Work as fellow intern in the subject of computer science in the degree of mathematics

Volunteering for building software in Sierra Leone

Lunsar, Sierra Leone

📅 January 2024 📍 Saint John of God hospital

Project organized by a cooperation association inside Polytechnic University of Catalonia. Building software for a hospital in Sierra Leone.

PROGRAMMING LANGUAGES

- Lean
- Python, C++/C#, $\text{\LaTeX}$
- Macaulay2, Sage

SCHOLARSHIPS

Full Degree Scholarship

Polytechnic University of Catalonia (UPC)

📅 2020 – Today

📍 Barcelona

Full scholarship granted by the Interdisciplinary Higher Education Centre (CFIS) for the whole degree of Mathematics and Engineering Physics.

Full Residence Scholarship

Polytechnic University of Catalonia (UPC)

📅 2020 – Today

📍 Barcelona

Full residence scholarship granted by the Interdisciplinary Higher Education Centre (CFIS) for the whole degree.

AWARDS



Gold Medal in Spain's Physics Olympiad

National competition held annually in Spain for pre-university students



Silver Medal in Ibero-American Physics Olympiad

International competition held annually for pre-university students



Classification to Spain's Physics Olympiad

National competition held annually in Spain for pre-university students

MISC

Graphic Design

- Inkscape, Gimp, 3D-Design

Music

- Professional studies in piano and trumpet

LANGUAGES

English TOEFL 103

Spanish Native

Catalan Advanced

German Beginner

HOBBIES

- Music, jamming sessions and dancing
- Sports (running, swimming, hiking, tennis, ...)
- Reading essays (history, philosophy, music, ...)
- Drawing in charcoal

REFEREES

Prof. Àlvarez, Josep

✉ josep.alvarez@upc.edu

📍 Polytechnic University of Catalonia (UPC)

Professor of commutative algebra.

Prof. van Doorn, Floris

✉ vdoorn@math.uni-bonn.de

📍 University of Bonn

Professor of formalization.

Prof. Rué, Juan José

✉ juan.jose.rue@upc.edu

📍 Polytechnic University of Catalonia (UPC)

Director of Interdisciplinary Higher Education Centre (CFIS).

Prof. Barja, Miguel Ángel

✉ miguel.angel.barja@upc.edu

📍 Polytechnic University of Catalonia (UPC)

Professor of algebraic geometry and former director of Interdisciplinary Higher Education Centre (CFIS).

CONFERENCES

ItaLean

University of Bologna

📅 December 2025

📍 Bologna, Italy

Southwest Local Algebra Meeting

Arizona State University

📅 March 2025

📍 Phoenix, Arizona

Hypatia graduate school

Mathematical Research Centre (CRM)

📅 June 2024

📍 Aut. Univ. of Barcelona

Jae School

Institute of Mathematical Sciences (ICMAT)

📅 July 2023

📍 Aut. Univ. of Madrid

Conferences at CRM

Mathematical Research Centre (CRM)

📅 July 2022

📍 University of Barcelona